A tetszőleges állapottér

$S_2, S \subseteq A \times (\bar{A} \cup \{fail\})^{**}$ program

$F \subseteq A \times A$ feladat

S megoldja F-et $\longrightarrow$ $D_F \subseteq D_{p(S)}$

$\forall a \in D_F : p(S)(a) \subseteq F(a)$

$S \subseteq S_2$

$$\boxed{S_2 \text{ program is megoldja } F\text{-et?}}$$

$A = \{1, 2\}$

$S = \{1 \to \langle 1 \rangle, 2 \to \langle 2 \rangle\}$

$S_2 = \{1 \to \langle 1 \rangle, 2 \to \langle 2 \rangle, 1 \to \langle 1, fail \rangle, 2 \to \langle 2, fail \rangle\}$

$D_F \subseteq D_{p(S)}$

$M \subseteq A \times (\bar{A} \cup \{fail\})^{**}$, M-ne teljesüljön: $S_2 = S \cup M$.

Tegyük fel, hogy $\forall a \in A : M(a) = \{\langle a, fail \rangle\}$

$D_F \subseteq D_{p(S_2)}$ $\to$ általánosan nem igaz

A tetszőleges halmazon

$S \subseteq A \times (\bar{A} \cup \{fail\})^{**}$ program

$F \subseteq A \times A$ feladat

S megoldja F-et $\longrightarrow$ $D_F \subseteq D_{p(S)}$

$\forall a \in D_F : p(S)(a) \subseteq F(a)$

$\boxed{\text{Ha } F \text{ nem determinisztikus, akkor } S \text{ sem?}}$

$A = \{1, 2\}$

$F = \{(1,2), (2,1), (1,1)\}$

$S = \left\{ \begin{array}{l} 1 \to \langle 1,2 \rangle \\ 2 \to \langle 2,1 \rangle \end{array} \right\}$

$2 \cancel{\boxtimes} 1$ $\quad$ $1 \cancel{\boxtimes} 2$

$1 \boxtimes$

$\forall a \in D_F : \left(1 \leq |p(S)(a)| \leq |F(a)| \wedge p(S)(a) \subseteq F(a) \right)$

Elemi programok

A tetsz. állapotter

SKIP	$\forall a \in A : SKIP(a) = \{\langle a \rangle\}$
ABORT	$\forall a \in A : ABORT(a) = \{\langle a, fail \rangle\}$

S A feletti program.

$$\forall a \in A : S(a) \subseteq \{\langle a \rangle, \langle a, fail \rangle, \langle a, a, a, \ldots \rangle, \langle a, b \rangle \mid b \in A\}$$

S elemi program A felett.

Ekvivalens programok

$S_1 \subseteq A \times (\bar{A} \cup \{fail\})^{**}$ program

$S_2 \subseteq A \times (\bar{A} \cup \{fail\})^{**}$ program

$p(S_1) = p(S_2) \rightarrow S_1$ és S_2 ekvivalensek

Mely feladatot/feladatokat oldja meg az ABORT program egy betűt. A állapotban futtt?

$\boxed{D_F \subseteq D_P(ABORT) =} \emptyset$

$\emptyset \subseteq \emptyset$

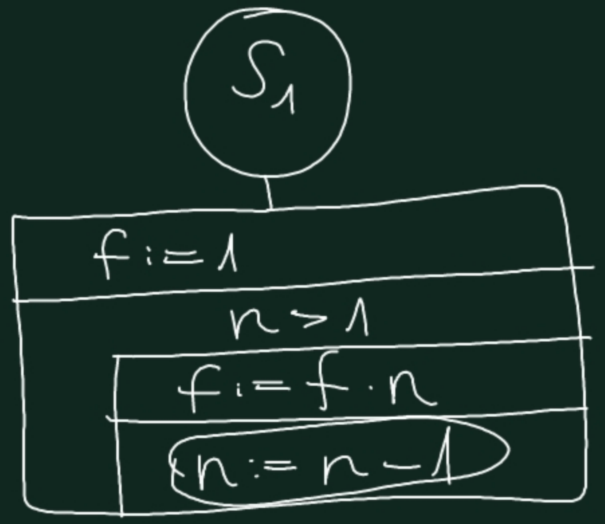
$D_F = \emptyset$

$F = \emptyset$

Csak az üres feladatot.

$A = (n : \mathbb{Z}, f : \mathbb{Z})$

$$\boxed{\varphi(S_1) = \left\{ \left(\begin{matrix} \{n : \alpha, f : \beta\}, \\ \{n : \alpha, f : 1\} \end{matrix} \right) \in A \times A \;/\; \alpha \leq 1 \right\}}$$



Hat írozzuk meg $\varphi(S_1)$-et!

Kezdőállapot

$$\{n : \alpha, f : \beta\} \in A$$

$$\{n : \alpha, f : 1\} \in A$$

$\alpha \leq 1$

$$\{n : \alpha, f : 1\}$$

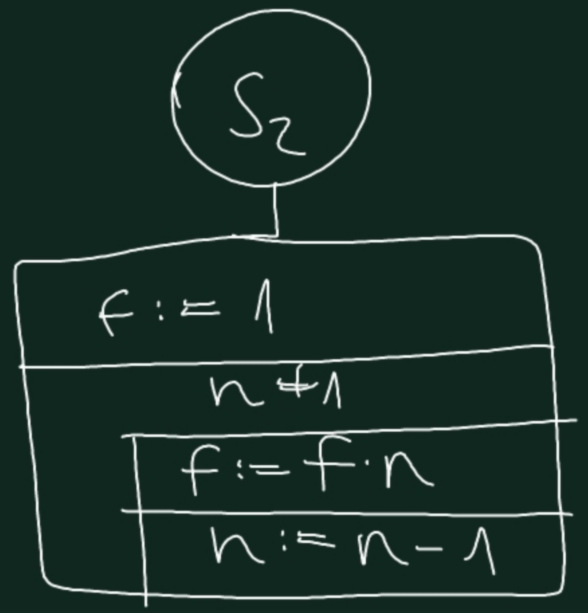
$\alpha > 1$

$$\{n : 1, f : \alpha!\}$$

$1 \cdot \alpha \cdot (\alpha - 1) \cdot (\alpha - 2) \cdot \ldots \cdot 2 = \alpha!$

$$\left\{ \left(\{n : \alpha, f : \beta\}, \{n : 1, f : \alpha!\} \right) \in A \times A \;\middle|\; \alpha > 1 \right\}$$

$A = (n : \mathbb{R}, f : \mathbb{R})$



kezdőállapot

$\{n : \alpha, f : \beta\} \in A$

$\{n : \alpha, f : 1\} \in A$

$\alpha = 1$

$\alpha > 1$

$\alpha \neq 1$

$\alpha < 1$

$\{n : \alpha, f : 1\}$

nincs terminálás

$\{n : 1, f : \alpha!\}$

unió

$p(S)$

$\{(\{n : 1, f : \beta\}, \{n : 1, f : 1\}) \in A \times A\} \subseteq p(S_2)$

$\{(\{n : \alpha, f : \beta\}, \{n : 1, f : \alpha!\}) \in A \times A / \alpha > 1\} \subseteq p(S_2)$

$$D_{P(S_1)} = A$$

$$D_{P(S_2)} = \{\{n:\alpha, f:\beta\} \in A \mid \alpha \geq 1\}$$

Ekvivalens-e S_1 és S_2 program?

$$D_{P(S_1)} \neq D_{P(S_2)}$$

<u>Nem.</u>

$$A = (n : \mathbb{Z}, f : \mathbb{Z})$$

$$F = \{ (a, b) \in A \times A \,/\, n(a) > 0 \wedge f(b) = n(a)! \}$$

S_1 és S_2 megoldja-e az F feladatot?

$$D_F = \{ a \in A \,/\, n(a) > 0 \} \subseteq D_{p(S_1)} = A \checkmark \text{igaz}$$

$$D_F \subseteq D_{p(S_1)}$$
$$D_F \subseteq D_{p(S_2)}$$

$$D_F = \{ a \in A \,/\, n(a) > 0 \} \subseteq D_{p(S_2)} = \{ a \in A \,/\, n(a) \geq 1 \} \checkmark \text{igaz}$$

$$\forall a \in D_F : p(S_1)(a) \subseteq F(a)$$
$$\forall a \in D_F : p(S_2)(a) \subseteq F(a)$$

$a \in D_F$, error...

$$F(a) = \{ b \in A / F(b) = n(a)! \}$$

$$p(S_1)(a) = \{ b \in A / F(b) = n(a)! \}$$

$$p(S_2)(a) = \{ b \in A / F(b) = n(a)! \}$$

$$\forall a \in D_F : \quad F(a) = p(S_1)(a) = p(S_2)(a)$$

S_1, S_2 is megoldja F-et.

A tetszőleges állapottér

S_1, S_2 A feltételprogramok

$S_1 \subseteq S_2$

(a) $D_p(S_1) \subseteq D_p(S_2)$ teljesül? Nem

(b) $D_{p(S_2)} \subseteq D_{p(S_1)}$ teljesül? Igen

$S_2 = S_1 \cup M$, ahol $M \subseteq A \times (\bar{A} \cup \{Fail\})^{**}$

$$D_p(S_2) = \{a \in A \mid (S_1 \cup M)(a) \subseteq \bar{A}^*\} =$$

$$= \{a \in A \mid S_1(a) \subseteq \bar{A}^* \wedge M(a) \subseteq \bar{A}^*\}$$

$$D_p(S_1) = \{a \in A \mid S_1(a) \subseteq \bar{A}^*\}$$

A tetszőleges állapottér

S_1, S_2, F A feltételi programok / feladat

$S_2 \subseteq S_1$

S_1 megoldja F-et $\longrightarrow$ $D_F \subseteq D_{p(S_1)}$

$\forall a \in D_F : p(S_1)(a) \subseteq F(a)$

$\forall a \in A : S_2(a) \subseteq S_1(a)$

$\forall a \in (D_{p(S_1)} \cap D_{p(S_2)}) : p(S_2)(a) \subseteq p(S_1)(a)$

S_2 megoldja F-et?

$D_F \subseteq D_{p(S_1)} \subseteq D_{p(S_2)}$ ✓

$\forall a \in D_F : p(S_2)(a) \subseteq p(S_1)(a) \subseteq F(a)$ ✓

Igen.